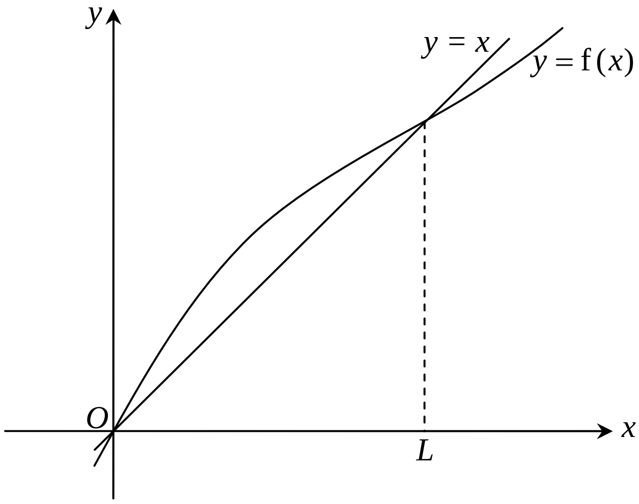
1 A sequence of positive real numbers $x_1, x_2, x_3, \ldots$ satisfies the recurrence relation

$$x_{n+1} = \frac{1}{2}x_n + \frac{x_n}{e^{x_n}}, \text{ for } n \geq 1.$$

(a) Prove algebraically that, if the sequence $\{ x_n \}$ converges, then it converges to L, a real number to be determined exactly. [2]

The diagram below shows the graphs of $y = x$ and $y = \text{f}(x) = \frac{1}{2}x + \frac{x}{e^x}$ sketched on the same axes.



(b) Given that $0 < x_1 < L$, make use of the sketch provided to demonstrate how the recurrence relation $x_{n+1} = \text{f}(x_n)$ will produce a sequence $x_1, x_2, x_3, \ldots$ converging to L. [2]

2 **(i)** Expand $\left(2 + \frac{1}{x}\right)^{-2}$ in ascending powers of x, up to and including the terms in x^5. State the range of x for which the expansion is valid. [4]

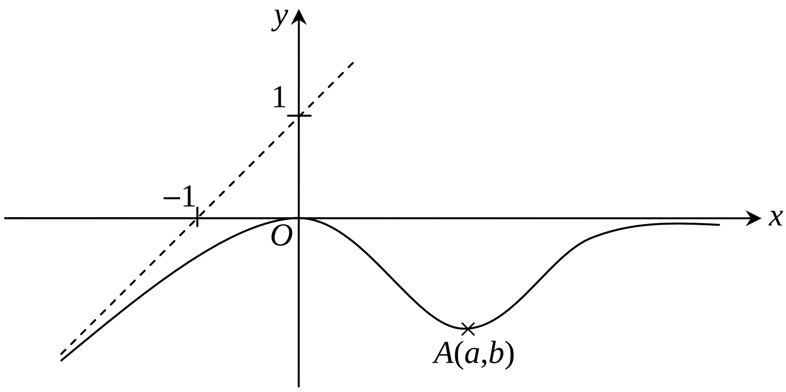
(ii) Write down, in its simplest form, the coefficient of x^r of the expansion in part **(i)**. [1]

3 The complex number w is given by $w = \dfrac{\left(\cos\frac{\pi}{3} + \text{i}\sin\frac{\pi}{3}\right)^4}{\left(\cos\frac{\pi}{6} - \text{i}\sin\frac{\pi}{6}\right)^5}$.

(i) Find $|w|$ and the exact value of arg(w). Hence, find the value of w^6. [3]

(ii) Show that $(\cos\theta + \text{i}\sin\theta)(1 + \cos\theta - \text{i}\sin\theta) = 1 + \cos\theta + \text{i}\sin\theta$. Hence, find the value of $(1+w)^6 + (1+w^*)^6$, where w^* is the complex conjugate of w. [3]

4 The diagram shows the graph of $y = \mathrm{f}(x)$. The origin O and the point $A(a\ ,\ b)$ are the two stationary points of the curve. The straight line $y = x + 1$ and the x-axis are the two asymptotes of the curve.



Sketch, on separate clearly labeled diagrams, the graphs of

(i) $y = \mathrm{f}(x + a) - b$, [3]

(ii) $y = \mathrm{f}'(x)$, where $\mathrm{f}'(x)$ denotes the first derivative of $\mathrm{f}(x)$. [2]

In each case, you are also to indicate clearly all the asymptotes and the coordinates of the points corresponding to O and A.

5 **(a)** **(i)** Show that $\dfrac{\mathrm{d}}{\mathrm{d}x}\left(\cos^{-1}\left(\dfrac{2x}{1+x^2}\right)\right) = \dfrac{-2}{1+x^2}$ for $-1 \le x \le 1$.

Explain, in your working, how you use the condition $-1 \le x \le 1$ to conclude your answer. [3]

(ii) Hence, determine the constants A and B satisfying the equation

$$\cos^{-1}\left(\frac{2x}{1+x^2}\right) + A\tan^{-1}x = B, \text{ for } -1 \le x \le 1.$$ [3]

(b) A right cone with base area A has a fixed height of 10 cm. Given that the base area is increasing at a rate of 2 $\mathrm{cm^2 s^{-1}}$, calculate the rate of change of the volume of the cone. [2]

6 In a chemical reaction, the amount of substance X and substance Y are being observed. There are x grams of substance X and y grams of substance Y at time t minutes. Using a mathematical model, the variable x and y satisfy the equations

$$\frac{\mathrm{d}y}{\mathrm{d}t} = 2 - x \quad \text{and} \quad \frac{\mathrm{d}x}{\mathrm{d}t} = e^x, \text{ for } 0 \le t < 1.$$

(i) Initially, there are no traces of substance X and substance Y present in the reaction. Express y in terms of x. [4]

(ii) The amount of substance Y during the reaction is the maximum at time k minutes. Find the value of k. [4]

7 A curve C is defined by the parametric equations $x = t^2$ and $y = \frac{1}{t}$, $t \neq 0$.

(i) Show that the equation of the tangent to the curve at the point $A\left(t^2, \frac{1}{t}\right)$ is

$$2t^3y - 3t^2 + x = 0.$$ [3]

(ii) Sketch the curve C and indicate all the line segments emitting from $B(12,0)$ that are tangential to the curve C. [2]

(iii) Find the area of the region R bounded by the curve C, the line $x = a$ (where $a > 12$) and the line segments mentioned in part **(ii)**. [4]

8 The functions f and g are defined as follows :

$$\mathrm{f} : x \to \ln(x+2)\ ,\ x \geq -2\ ,$$
$$\mathrm{g} : x \to \mathrm{f}(|x|)\quad ,\ x \in \Re\ .$$

(a) Solve for all the values of x satisfying the equation fg(x) = gf(x). [4]

(b) **(i)** Briefly explain why the inverse function of g cannot be formed. [1]

(ii) The function g has an inverse if its domain is restricted to $x \leq a$. Determine the largest value of a. [1]

(iii) Define completely the inverse function g^{-1} corresponding to the largest value of a determined in part **(b)(ii)**. [3]

9 The lines l_1 and l_2 have equations

$$\mathbf{r} = \begin{pmatrix} 0 \\ 3 \\ 1 \end{pmatrix} + \lambda \begin{pmatrix} 1 \\ -2 \\ 1 \end{pmatrix} \text{ and } \mathbf{r} = \begin{pmatrix} 1 \\ 0 \\ 2 \end{pmatrix} + \mu \begin{pmatrix} -2 \\ 4 \\ -2 \end{pmatrix}$$

respectively, where λ and μ are real parameters.

(a) Show that the lines l_1 and l_2 are distinct parallel lines. [2]

(b) The points A and B have coordinates (0, 3, 1) and (1, 0, 2) respectively. The point P is a point on the line l_1 such that the angle APB is a right angle.

(i) Find the coordinates of the point P. [4]

(ii) Hence, find the shortest distance between the lines l_1 and l_2. [2]

(c) Find the vector equation, in scalar product form, of the plane that contains the lines l_1 and l_2. [2]

10 The curve C has the equation $y = \dfrac{ax^2 - x + b}{x + c}$, $x \neq -c$, where a, b and c are real constants.

(a) Write down the equations of all the asymptotes of the curve. [3]

(b) In the case when $a = -1$ and $c = 1$, the curve has a negative gradient throughout the range of x for which it is defined.

(i) Deduce the range of b. [4]

(ii) Sketch the curve C when b takes the least positive integral value. [2]

11 **(a)** An arithmetic progression **A** has first term 1 and common difference 2. A geometric progression **G** has first term a and the sum of the first n terms of **G** may be given by $\frac{a}{2}\left(3^n - 1\right)$, where a is an integer.

(i) Given that every term in **G** is also a term in **A**, find the condition to be satisfied by a. [3]

(ii) Prove that the progression

$$2^{t_1},\ 2^{t_2},\ 2^{t_3},\ 2^{t_4},\ \ldots$$

is a geometric progression, where t_1, t_2, t_3, t_4 are consecutive terms in **A**. [3]

(b) The cost price of a new machine is always \$70000. However, the value of the machine depreciates at a rate of 10% per annum, when in use. The output by this machine generates revenue of \$14000 in the first year, but the revenue generated in each subsequent year shows a decrease of 7%.

(i) Find the value of the machine after 5 years. [2]

(ii) Calculate the total revenue that could possibly be generated by the machine. [2]

(iii) For economic reasons, the machine has to be replaced by a new one when the sum of the total revenue received and the prevailing value of the old machine exceed the cost of a new machine by at least \$40000. Calculate the number of years the machine is in operation (including the year that the machine is being replaced). [3]

12 **(a)** **(i)** Find $\int \frac{x}{\sqrt{x-1}}\,\mathrm{d}x$, simplifying your answer as much as possible. [3]

(ii) Show that $\frac{4}{x^2(x-2)} = \frac{1}{x-2} - \frac{1}{x} - \frac{2}{x^2}$. [1]

Use the substitution $u = 2 + e^{-x}$ to evaluate the exact value of

$$\int_0^{\ln 4} \frac{4}{(2+e^{-x})^2}\,\mathrm{d}x.$$ [4]

(b) Region R is the area bounded by the curves $y = \sin x$, $y = \tan x$ and the line $y = 1$, where $0 \le x \le \frac{\pi}{2}$.

(i) Sketch the region of R, labeling all the vertices of the region clearly. [2]

(ii) Find the exact volume of the solid of revolution obtained when the region R is rotated through 2π radians about the x-axis. [4]

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